Livrable 2 : Modélisation conceptuelle et logique de la base ArtConnect

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Bases de données 2 : Sécurité et Automatisation

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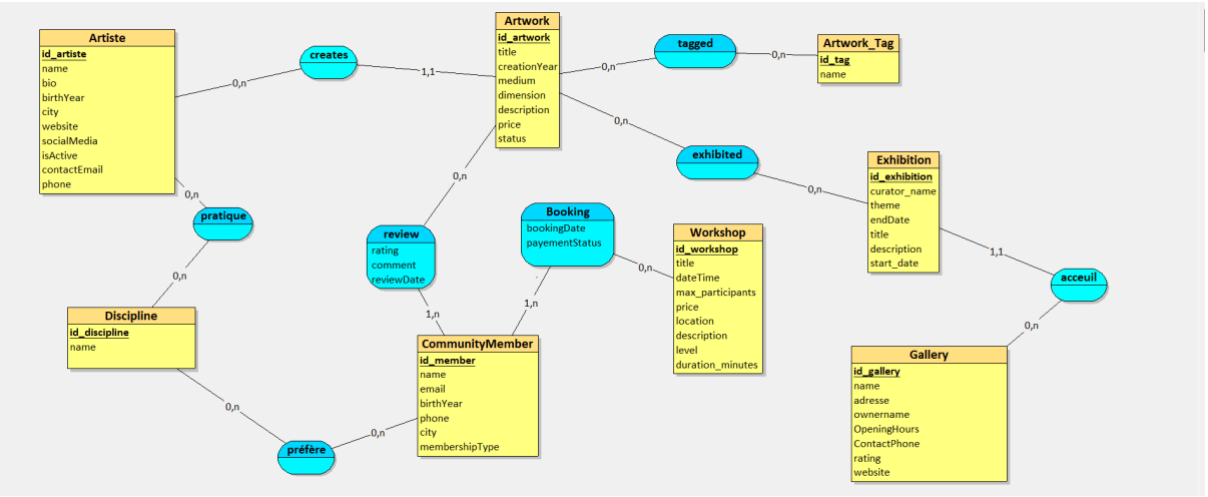
1. Identifier les entité et les interactions

Entité	Attributs	Identifiant	Dépendance clé
Artist	name, bio, birth_year, contactEmail, phone, city, website, socialMedia, isActive, id_artwork(FK)	id_artist (PK auto)	id_artist → tous les attributs
Artwork	title, creationYear, type, medium, dimension, description, price, statue	id_artwork (PK auto)	id_artwork → tous
Gallery	name, address, owner_name, opening_hours, contact_phone, rating, website	id_gallery (PK auto)	id_gallery → tous
Exhibition	title, start_date, end_date, description, curator_name, theme, id_gallery (FK)	id_exhibition (PK auto)	id_exhibition → tous
Workshop	title, date_time, duration_minutes, max_participants, price, location, description, level, id_instructor (FK)	id_workshop (PK auto)	id_workshop → tous
CommunityMember	name, email, birth_year, phone, city, membership_type	id_member (PK auto)	id_member → tous
Discipline	name	id_discipline (PK auto)	id_discipline → name
ArtworkTag	name	id_tag (PK auto)	id_tag → name
Booking	booking_date, payment_status, id_workshop (FK), id_member (FK)	id_booking (PK auto)	id_booking → tous
Review	rating, comment, review_date, id_reviewer (FK), id_artwork (FK)	id_review (PK auto)	id_review → tous

2. Créer un modèle conceptuel des données (MCD)



3. Créer un modèle logique des donées (MLD)

Artiste = (id_artiste INT, name VARCHAR(50), bio VARCHAR(50), birthYear INT, city VARCHAR(100), website VARCHAR(300), socialMedia VARCHAR(300), isActive LOGICAL, contactEmail VARCHAR(200), phone VARCHAR(30));
Discipline = (id_discipline INT, name VARCHAR(100));
CommunityMember = (id_member INT, name VARCHAR(150), email VARCHAR(200), birthYear INT, phone VARCHAR(30), city VARCHAR(100), membershipType ENUM('free', 'premium'));
Artwork = (id_artwork INT, title VARCHAR(200), creationYear INT, medium VARCHAR(100), dimension VARCHAR(300), description TEXT, price DECIMAL(12,2), status ENUM('FOR_SALE', 'SOLD', 'EXHIBITED'), #id_artiste);
Gallery = (id_gallery INT, name VARCHAR(200), adresse VARCHAR(50), ownername VARCHAR(50), OpeningHours VARCHAR(50), ContactPhone VARCHAR(50), rating DECIMAL(15,2), website VARCHAR(50));
Workshop = (id_workshop INT, title VARCHAR(50), dateTime DATETIME, max_participants INT, price DECIMAL(15,2), location VARCHAR(50), description TEXT, level ENUM('beginner', 'intermediate', 'advanced'), duration_minutes INT);
Artwork_Tag = (id_tag INT, name VARCHAR(50));
Exhibition = (id_exhibition INT, curator_name VARCHAR(150), theme VARCHAR(200), endDate DATE, title VARCHAR(200), description TEXT, start_date DATE, #id_gallery);

exhibited = (#id_artwork, #id_exhibition);
tagged = (#id_artwork, #id_tag);
pratique = (#id_artiste, #id_discipline);
prefere = (#id_discipline, #id_member);
review = (#id_artwork, #id_member, rating INT, comment VARCHAR(150), reviewDate DATE);
Booking = (#id_member, #id_workshop, bookingDate DATE, paymentStatus ENUM(‘Pending’, ‘Paid’, ‘Cancelled’)

1FN : il faut que chaque attribut soit atomique

openingHours dans Gallery stocke plusieurs horaires dans un seul champ
dimensions dans Artwork stocke largeur, hauteur et profondeur dans un seul champ.
socialMedia dans Artiste pourrait stocker plusieurs liens en un seul champ.

Voici notre base de donnée 1FN :

Artiste = (id_artiste INT, name VARCHAR(50), bio TEXT, birthYear INT, city VARCHAR(100), website VARCHAR(300), isActive BOOLEAN, contactEmail VARCHAR(200), phone VARCHAR(30))
ArtisteSocial = (id_social INT, platform VARCHAR(50), url VARCHAR(300), #id_artiste)
Gallery = (id_gallery INT, name VARCHAR(200), address VARCHAR(200), ownerName VARCHAR(100), contactPhone VARCHAR(30), rating DECIMAL(3,2), website VARCHAR(200))
GalleryHours = (id_hours INT, dayOfWeek ENUM('MON','TUE','WED','THU','FRI','SAT','SUN'), openTime TIME, closeTime TIME, #id_gallery)
Artwork = (id_artwork INT, title VARCHAR(200), creationYear INT, medium VARCHAR(100), width DECIMAL(8,2), height DECIMAL(8,2), depth DECIMAL(8,2), description TEXT, price DECIMAL(12,2), status ENUM('FOR_SALE','SOLD','EXHIBITED'), #id_artiste)
Discipline = (id_discipline INT, name VARCHAR(100))
Artwork_Tag = (id_tag INT, name VARCHAR(50))
CommunityMember = (id_member INT, name VARCHAR(150), email VARCHAR(200), birthYear INT, phone VARCHAR(30), city VARCHAR(100), membershipType ENUM('free','premium'))
Workshop = (id_workshop INT, title VARCHAR(100), dateTime DATETIME, max_participants INT, price DECIMAL(15,2), duration_minutes INT, description TEXT, level ENUM('beginner','intermediate','advanced'), location VARCHAR(200))
Exhibition = (id_exhibition INT, title VARCHAR(200), description TEXT, startDate DATE, endDate DATE, curatorName VARCHAR(150), theme VARCHAR(200), #id_gallery)

exhibited = (#id_artwork, #id_exhibition)
tagged = (#id_artwork, #id_tag)
pratique = (#id_artiste, #id_discipline)
prefere = (#id_discipline, #id_member)
review = (#id_artwork, #id_member, rating INT, comment VARCHAR(150), reviewDate DATE)
Booking = (#id_member, #id_workshop, bookingDate DATE, paymentStatus ENUM(‘Pending’, ‘Paid’, ‘Cancelled’)

2FN : Pas de dépendance partielle

Aucun attribut à une dépendance partielle donc la base de donnée est déjà en 2FN

3FN : aucun attribut non-clé ne dépende d'un autre attribut non-clé (aucune dépendance transitive)

Location dans workshop est un champ libre donc si deux workshop a lieu au même endroit alors l'adresse est dupliqué donc on extrait une table Location.
Même raisonnement pour gallery. On peut mutualisé avec celle de workshop
City dans artist et community_member est une valeur libre. On crée un table city pour les deux.

Voici la base de donnée en 3FN :

Artiste = (id_artiste INT, name VARCHAR(50), bio TEXT, birthYear INT, website VARCHAR(300), isActive BOOLEAN, contactEmail VARCHAR(200), phone VARCHAR(30),#id_city)
ArtisteSocial = (id_social INT, platform VARCHAR(50), url VARCHAR(300), #id_artiste)
Gallery = (id_gallery INT, name VARCHAR(200), ownerName VARCHAR(100), contactPhone VARCHAR(30), rating DECIMAL(3,2), website VARCHAR(200), #id_location)
GalleryHours = (id_hours INT, dayOfWeek ENUM('MON','TUE','WED','THU','FRI','SAT','SUN'), openTime TIME, closeTime TIME, #id_gallery)
Artwork = (id_artwork INT, title VARCHAR(200), creationYear INT, medium VARCHAR(100), width DECIMAL(8,2), height DECIMAL(8,2), depth DECIMAL(8,2), description TEXT, price DECIMAL(12,2), status ENUM('FOR_SALE','SOLD','EXHIBITED'), #id_artiste)
Discipline = (id_discipline INT, name VARCHAR(100))
Artwork_Tag = (id_tag INT, name VARCHAR(50))
CommunityMember = (id_member INT, name VARCHAR(150), email VARCHAR(200), birthYear INT, phone VARCHAR(30), membershipType ENUM('free','premium'), #id_city)
Workshop = (id_workshop INT, title VARCHAR(100), dateTime DATETIME, max_participants INT, price DECIMAL(15,2), duration_minutes INT, description TEXT, level ENUM('beginner','intermediate','advanced'),#id_location)
Exhibition = (id_exhibition INT, title VARCHAR(200), description TEXT, startDate DATE, endDate DATE, curatorName VARCHAR(150), theme VARCHAR(200), #id_gallery)
City = (id_city INT, name VARCHAR(100), region VARCHAR(100), country VARCHAR(100))
Location = (id_location INT, name VARCHAR(100), address VARCHAR(200),

#id_city)

exhibited = (#id_artwork, #id_exhibition)
tagged = (#id_artwork, #id_tag)
pratique = (#id_artiste, #id_discipline)
prefere = (#id_discipline, #id_member)
review = (#id_artwork, #id_member, rating INT, comment VARCHAR(150), reviewDate DATE)
booking = (#id_member, #id_workshop, bookingDate DATE, paymentStatus ENUM('Pending', 'Paid', 'Cancelled')

4. Créer le schéma de la base de données

Le script MySQL se trouve dans le dossier Livrable2 sur GitHub.